Table 21. Initial architecture decision records

ADR	Decision	Rationale	Alternatives considered
001	Modular services with open interfaces	Components can be replaced without rewriting others; avoids lock-in	Monolithic application; single vendor suite
002	Retrieval before generation	Answers stay traceable to official text; reduces invented content	Pure generative assistant; fine-tuned model without retrieval
003	Open-weight models hosted in Bangladesh	Data residency; control over versions; no data sent abroad	Commercial API models for all data classes
004	Hybrid lexical and dense retrieval	Lexical search is transparent and strong on exact terms; dense search handles paraphrase	Dense only; lexical only
005	Append-only, hash-chained audit log	Independent verification of what the system did	Ordinary application logs
006	Containers and infrastructure as code	Reproducible, auditable environments	Manually configured servers
007	Accessibility and low bandwidth first	Reach users outside Dhaka and on mobile	Desktop-first interface
008	Public data first (P0)	Value without privacy risk while governance matures	Start with administrative records

Source: BAPIL Technical Blueprint, Version 1.0 (2026). CC BY 4.0.